

SOCORRO, NM, USA

WMO: 723620

Lat: 34.022N

Lon: 106.903W

Elev: 1486

StdP: 84.70

Time Zone: -7.00 (NAM)

Period: 09-19

WBAN: 93040

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.3	(c) -5.9	(d) -20.2	(e) 0.7	(f) 4.1	(g) -18.1	(h) 0.9	(i) 6.0	(j) 12.3	(k) 7.4	(l) 11.0	(m) 7.5	(n) 3.0	(o) 330	(p) 0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	17.2	37.3	15.6	36.0	15.5	34.8	15.5	18.9	26.8	18.4	26.6	17.9	26.8	4.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	14.7	20.7	16.3	13.9	20.6	15.5	13.2	20.6	60.2	26.8	58.4	26.1	56.7	26.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.1	8.0	-12.5	39.4	5.0	0.9	-16.1	40.1	-19.1	40.6	-21.9	41.1	-25.5	41.8
			-13.4	20.0	4.6	0.8	-16.7	20.6	-19.5	21.1	-22.1	21.5	-25.4	22.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.0	7.5	12.2	15.8	19.5	26.4	26.7	25.4	22.5	16.2	9.1	4.1		
	DBStd	8.80	3.60	4.70	3.72	3.74	3.85	2.45	1.99	1.95	2.69	3.64	3.89	4.31		
	HDD10.0	556	187	92	21	4	1	0	0	0	0	4	62	185		
	HDD18.3	1866	444	304	191	91	33	0	0	0	3	83	276	441		
	CDD10.0	2681	2	21	89	178	294	492	518	478	374	196	36	3		
	CDD18.3	949	0	0	1	15	68	242	259	220	127	17	0	0		
	CDH23.3	11921	0	4	96	363	1113	3371	2991	2352	1329	297	4	0		
	CDH26.7	5952	0	0	10	79	452	1961	1623	1187	576	64	0	0		
(14) Wind	WSAvg	3.3	3.1	3.3	3.7	4.1	4.0	3.5	3.0	2.8	3.0	3.0	3.2	3.1		
(15) Precipitation	PrecAvg	236	9	9	9	9	11	15	38	49	37	28	12	16		
	PrecMax	355	46	36	54	70	42	75	98	161	123	136	41	59		
	PrecMin	115	0	0	0	0	0	0	4	5	3	0	0	0		
	PrecStd	60	11	9	11	14	11	17	25	32	27	29	12	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	23.9	27.9	31.1	34.8	39.1	38.2	37.1	35.0	30.8	23.9	19.1		
		MCWB	6.8	8.9	10.5	11.4	13.3	15.4	16.2	16.6	15.9	13.7	9.8	7.8		
	2%	DB	16.0	21.6	26.1	28.9	32.9	37.8	36.7	35.3	33.3	28.3	22.1	16.7		
		MCWB	5.8	8.1	9.5	11.0	12.7	15.3	16.1	16.5	15.7	13.0	9.6	6.4		
	5%	DB	13.8	19.2	23.9	27.3	31.2	36.5	35.4	34.0	32.2	26.8	19.5	14.0		
		MCWB	4.7	6.9	8.6	10.4	12.1	14.9	16.2	16.5	15.5	12.5	8.5	5.5		
	10%	DB	11.4	16.8	21.7	25.6	29.3	35.1	34.0	32.6	30.4	24.9	17.1	11.5		
		MCWB	3.4	5.9	7.7	9.7	11.5	14.4	16.2	16.5	15.1	11.8	7.4	4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	9.8	11.1	12.9	15.1	17.8	19.1	19.8	19.1	17.3	11.7	8.9		
		MCDB	16.1	22.2	24.6	24.7	29.7	31.1	26.3	28.2	25.5	22.4	18.5	16.4		
	2%	WB	6.5	8.5	10.1	11.8	13.8	17.0	18.6	19.0	18.5	15.4	10.4	7.5		
		MCDB	13.7	19.7	24.0	24.4	27.9	30.6	26.5	27.6	25.2	21.8	17.8	14.7		
	5%	WB	5.3	7.4	9.1	11.0	12.8	16.4	18.2	18.5	17.7	14.1	9.4	6.1		
		MCDB	12.5	18.3	22.3	24.8	27.8	30.4	27.0	27.0	24.7	22.0	17.4	12.5		
	10%	WB	4.2	6.3	8.1	10.2	12.0	15.8	17.8	17.9	16.9	12.9	8.3	5.0		
		MCDB	10.4	15.5	20.5	24.1	26.8	30.2	27.4	26.8	24.6	22.0	16.5	10.8		
(35) Mean Daily Temperature Range	MDBR		13.8	15.4	16.8	16.8	16.8	17.2	14.7	14.6	14.7	15.5	14.8	13.0		
	5% DB	MCDBR	17.2	18.7	19.8	18.9	18.8	18.0	16.3	16.2	16.8	17.5	17.4	16.2		
		MCWBR	9.3	9.1	9.2	8.0	7.7	6.6	4.0	4.2	5.4	7.2	8.4	8.7		
	5% WB	MCDBR	14.9	17.9	18.5	17.6	17.3	16.3	14.0	14.0	12.9	14.3	15.3	14.2		
		MCWBR	8.2	8.8	8.6	7.3	6.8	5.2	3.1	3.6	4.1	6.0	7.8	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.240	0.246	0.267	0.289	0.306	0.327	0.368	0.344	0.312	0.275	0.253	0.243		
	taud		2.662	2.647	2.564	2.484	2.486	2.426	2.417	2.457	2.533	2.616	2.626	2.647		
	Ebn at Noon		1003	1024	1017	999	977	950	909	930	952	974	976	979		
	Edh at Noon		70	80	95	108	109	116	116	110	97	82	72	68		
(44) All-Sky Solar Radiation	RadAvg		3.41	4.40	5.76	7.04	7.79	7.88	6.98	6.46	5.80	4.78	3.71	3.04		
	RadStd		0.27	0.36	0.33	0.33	0.39	0.42	0.36	0.22	0.25	0.38	0.22	0.17		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+84	+50

Nomenclature: See separate page